Appendix B

ELEMENTS OF POINT SET TOPOLOGY

In the main text we assumed the familiarity with some basic notions of general (point set) topology that can be obtained in undergraduate-level education in mathematics. For self-containedness, the elementary concepts needed are presented in this Appendix, mainly based on the very accessible textbook of Henle [52].

B.1 CONTINUOUS TRANSFORMATIONS

We shall consider an Euclidean N-dimensional space E^N. The points of E^N are denoted by $X, Y, Z, \ldots$, and sets of points by $A, B, \ldots$. The distance between two points X and Y is denoted by $d(X, Y)$.[1]

Our first task is to develop a mathematically exact definition for a "continuous transformation." To this end, the most fundamental concept we need is the *neighborhood*:

Definition B.1 *The* δ-neighborhood *of a point* X, $N(X, \delta)$, *is defined as the set of points* Y *such that* $d(X, Y) < \delta$.

In E^2, δ-neighborhoods of X are open disks of radius δ centered at X, while in E^3 they are open balls.

Intuitively, the condition that a continuous transformation may not cause any splitting or tearing means that it must preserve the nearness of

[1] Topology can be developed without assuming a metric (a distance). Here we have chosen this more easily understandable approach adequate for our practical purposes. Of course, we shall be interested mainly in dimensions $N = 2, 3$.

points and sets: if a point is near some set, also the transformed point must be near the transformed set. To elaborate this intuitive notion, the concept of a neighborhood gives us a precise definition of nearness:

Definition B.2 *Point X is said to be* near *set A, if every neighborhood of X contains a point of A.*

Now we can define the fundamental concept of a continuous transformation:

Definition B.3 *A* continuous transformation *from a set D to a set R is a function f with domain D and range R such that for any point $X \in D$ and set $A \subset D$, if X is near A, then $f(X)$ is near the set $f(A) = \{f(Y)|Y \in A\}$.*

We further restrict the class of interesting transformations to *reversible* transformations, i.e., those transformations that have an inverse:

Definition B.4 *A* topological transformation *is a continuous transformation that has a continuous inverse. Two sets are said to be* topologically equivalent *if there is a topological transformation between them.*

Hence, for instance, the class of all shapes that can be formed from stretching an infinitely elastic ball without tearing or ripping it are topologically equivalent.

B.2 OTHER TOPOLOGICAL CONCEPTS

The concept of nearness is fundamental in that it forms the basis of defining other topological concepts. Other concepts used in the main text are defined as follows:

Definition B.5 *A set is* closed *if it contains all its near points. The* closure *of a set A, denoted by $c(A)$, is the set of points near A.*

Definition B.6 *A set is* bounded *if it is contained in a neighborhood.*

Definition B.7 *Sets that are both closed and bounded are said to be* compact.

Definition B.8 *A set A is* connected *if whenever A is divided into two disjoint, nonempty subsets B and C (i.e., whenever*

$$A = B \cup C,\ B \neq \emptyset,\ C \neq \emptyset,\ B \cap C = \emptyset$$

holds), one of these sets always contains a point near the other.

Definition B.9 *A set A is* open *if no point of A is near the complement of A.*

Definition B.10 *The* interior *of a set A, denoted by $i(A)$, is the set of points $\in A$ that are not near the complement of A.*

Definition B.11 *The* boundary *of a set A, $b(A)$, is the set of points that are near both A and its complement.*

B.3 TOPOLOGICAL SPACES

As seen in the previous section, all topological concepts were built on the notions of nearness and neighborhood. As nearness was defined in terms of neighborhood, we can actually study the topology of any sets by defining the concept of a neighborhood in a suitable way.

This general view of topology is based on the concept of a *topological space*, defined as follows:

Definition B.12 *A* topological space *is a set $\aleph$ with the choice of a class of subsets of $\aleph$ (each of which is called* neighborhood *of its points) such that each point of $\aleph$ is in some neighborhood, and the intersection of any two neighborhoods of a point contains a neighborhood of that point.*

Based on a particular instance of a topological space, all definitions of the previous sections can be carried out unaltered. For instance, the choice of $\aleph = E^N$ and neighborhoods $= \delta$-neighborhoods satisfies the two conditions above. Hence it leads to a topology on E^N. As this choice for a topology of E^N is intuitively the most natural one, it is called the *natural topology* of E^N.

It is possible to construct other kinds of topologies by selecting different neighborhoods satisfying the definition of a topological space. The definition of plane models in the main text is based on defining such a *special topology* on E^2. In that topology, neighborhoods are not always open disks as usually, but consist on two half-disks on identified edges. The reader is urged to verify that this definition really satisfies the notion of a topological space defined above.